

Shreeya Iyengar

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EDUCATION

Vanderbilt University, Nashville, TN

May 2029

Bachelor of Engineering, Electrical and Computer Engineering

Cumulative GPA: 3.55

Coursework: Java Programming, Physics 2: Electricity & Magnetism, Materials Science

Awards: V-START Summer Research Fellow, Ned McWherter Scholar, Scholastic Arts and Writing Medalist

Organizations: SCALE Microelectronics Student, Institute of Electrical and Electronic Engineers; Vanderbilt Engineering Student Ambassador; Society of Women Engineers; Physics Learning Assistant/Tutor; South Asian Cultural Exchange

WORK EXPERIENCE

Vanderbilt Institute of Surgery and Engineering: LOR3 Lab, Nashville, TN

Mar. 2026 — Present

Researcher

- Engineered advanced biomaterials for lung transplant applications using nanotechnology methods and contact angle testing
- Collaborated with industry partners and Vanderbilt Institute for Nanoscale Engineering to conduct nanoscale characterization and fabrication of coated silicon samples; cleanroom certified
- Built and maintained a flow circuit to quantify shear stress and material resilience; collected and analyzed experimental data for 30+ trials

Wond'ry IMPACT Program, Nashville, TN

Jan. 2026 — Present

Fellow

- Engaged with leaders in National Security, Healthcare and Sustainability to find unmet technical and human centered challenges
- Collaborated with a team to design a technology-driven product aimed at reducing healthcare access disparities
- Delivered a product pitch grounded in user research, design notes, and validated need-finding insights to over 15 business leaders

BRIDGES Youth Action Center, Memphis, TN

June 2022 — May 2025

Shelby County Youth District 4 Representative

- Coauthored a proposal for a county wide resolution requiring county commissioners to meet with their Youth Representative at least biyearly
- Invited to EdTrust's Day of Education Advocacy at TN Capitol Building; lobbied over six bills regarding quality of education during legislative appointments
- Organized and facilitated four forums for 200+ youth to discuss education, mental health and crime and build solutions with government officials

PERSONAL PROJECTS

Arduino Proximity Security Alert System

May 2026 - Present

- Engineered a sweeping radar-like mechanism that halts and triggers visual and audio alerts when objects are detected within a set range using Arduino IDE C++
- Implemented toggle-based motor control using button and joystick inputs with software debouncing and edge detection logic.
- Designed custom PCB using KiCAD based on breadboard prototype and subsequent schematic, soldered components together and designed 3D printed casing for durability and longtime use.

Autonomous Maze VEX Robot

Aug. 2023 — May 2025

Team Member

- Engineered and iteratively optimized a VEX robot to autonomously navigate a randomized obstacle maze, optimizing for size and accuracy based on Science Olympiad competition requirements.
- Designed external power source, built chassis within size constraints, and programmed VEX electronics including vision sensors and smart motors

LEADERSHIP EXPERIENCE

Multicultural Action Council, Nashville, TN

Sep. 2025 — Present

Board Member (2025-2026), First Year Relations Chair (2026-2027)

- Represented Multicultural Leadership Council (55+ cultural clubs) during club fairs and Vanderbilt admissions events; designed promotional signage and engage 100+ individuals
- Hosted collaborative cooking night with Vanderbilt Bengali Association, tracked event setup and attendance

SKILLS & INTERESTS

Technical Skills: Java, KiCAD, Hardware Prototyping and Debugging, PCB Design, Arduino IDE (C++), R, Fiji: ImageJ, OnShape, Canva, Microsoft Suite, Cleanroom Certified, Contact Angle Goniometer

Interests: Art, Cultural Advocacy, Fashion, Emerging Technologies, College Football